

Score Labspace

A learning experience for exploring and understanding the Score project and features.

Overview

👋 Welcome to the **Score Labspace**!

This is a learning experience for exploring and understanding the CNCF Sandbox project [Score](#) and its features.

You will be guided through instructions to run command lines with tools already pre-installed to better understand the concepts of Score and its features.

✅ Here is how this lab is structured:

- Introduction to Score
- Getting started with `score-compose`
- Getting started with `score-k8s`
- Advanced scenario with `score-compose`
- Advanced scenario with `score-k8s`
- Next steps

Let's Score it! 🎵

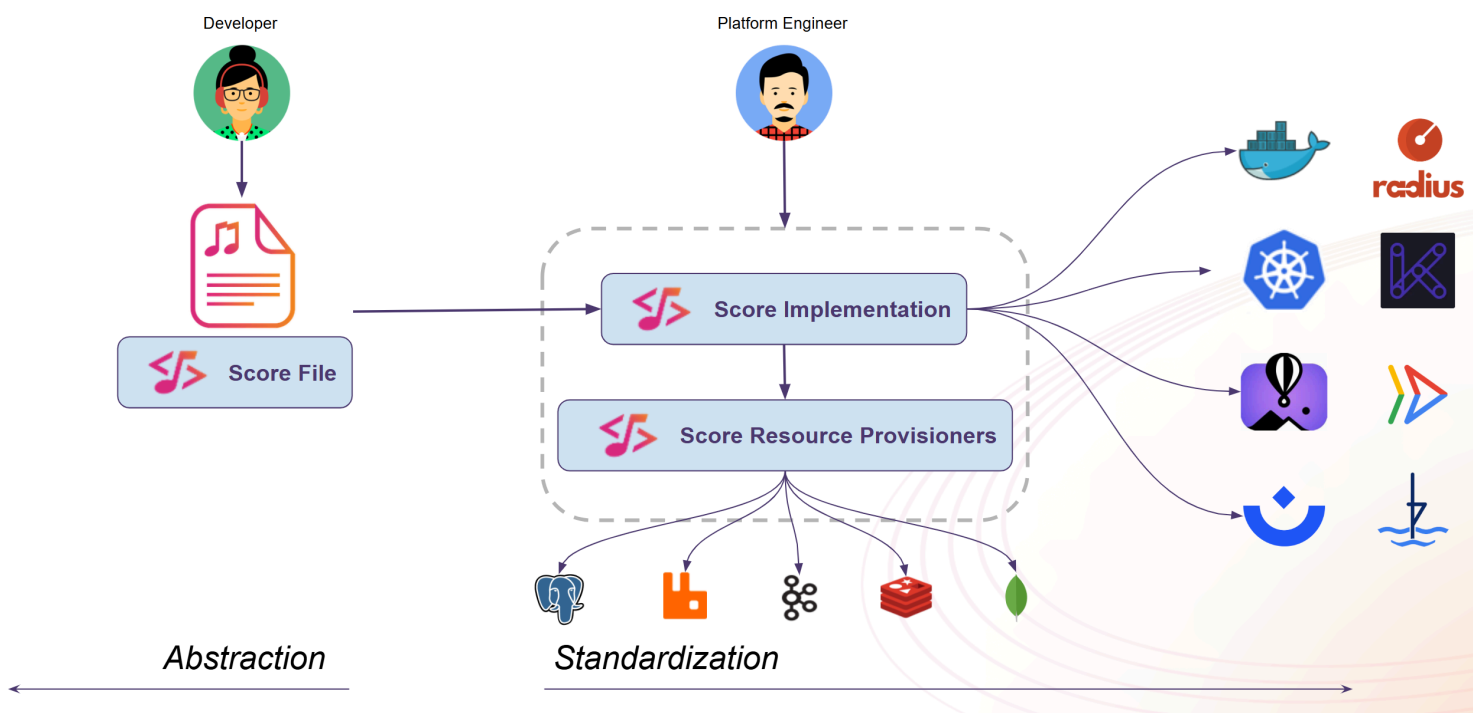
Enjoy! 🎉

What is Score?

Score is an open-source and a [CNCF Sandbox project](#).

Score aims to reduce developer toil and cognitive load by enabling the definition of a single file that works across multiple platforms in a vendor-neutral way, eliminating the need for tooling-specific syntax from platforms such as Docker or Kubernetes for example.

On one hand ("**Abstraction**"), the Score spec allows **Developers** to define how they want to deploy their Workloads. And on the other hand ("**Standardization**"), **Platform Engineers** define well supported golden paths by using Score implementations supported for their platform(s) with associated resource provisioners.



Score specification

First and foremost, Score is a Specification.

The Score Specification is what Developers will use to describe in a YAML file what their workload needs. Eventually someone later will deploy it somewhere with a Score implementation.

Simple Score file

The simplest Score file that you can have is:

```
apiVersion: score.dev/v1b1
metadata:
  name: my-workload
containers:
  my-container:
    image: .
```

This is defining that the workload `my-workload` has one container `my-container`, and at this stage the container image name is not known (`.`) and will be supplied later in the CI/CD workflow.

Note: it's not a Kubernetes Custom Resource.

More complex Score file

Now here is a more realistic workload definition, a public service talking to a PostgreSQL database ([score.yaml](#)):

```
apiVersion: score.dev/v1b1
metadata:
  name: hello-world
  annotations:
    tags: "nodejs,http,website,javascript"
containers:
  hello-world:
    image: .
    variables:
      PORT: "3000"
      MESSAGE: "Hello, World!"
      DB_DATABASE: ${resources.db.name}
      DB_USER: ${resources.db.username}
      DB_PASSWORD: ${resources.db.password}
      DB_HOST: ${resources.db.host}
```

```
    DB_PORT: ${resources.db.port}
resources:
  db:
    type: postgres
  dns:
    type: dns
  route:
    type: route
    params:
      host: ${resources.dns.host}
      path: /
      port: 8080
service:
  ports:
    www:
      port: 8080
      targetPort: 3000
```

Resources

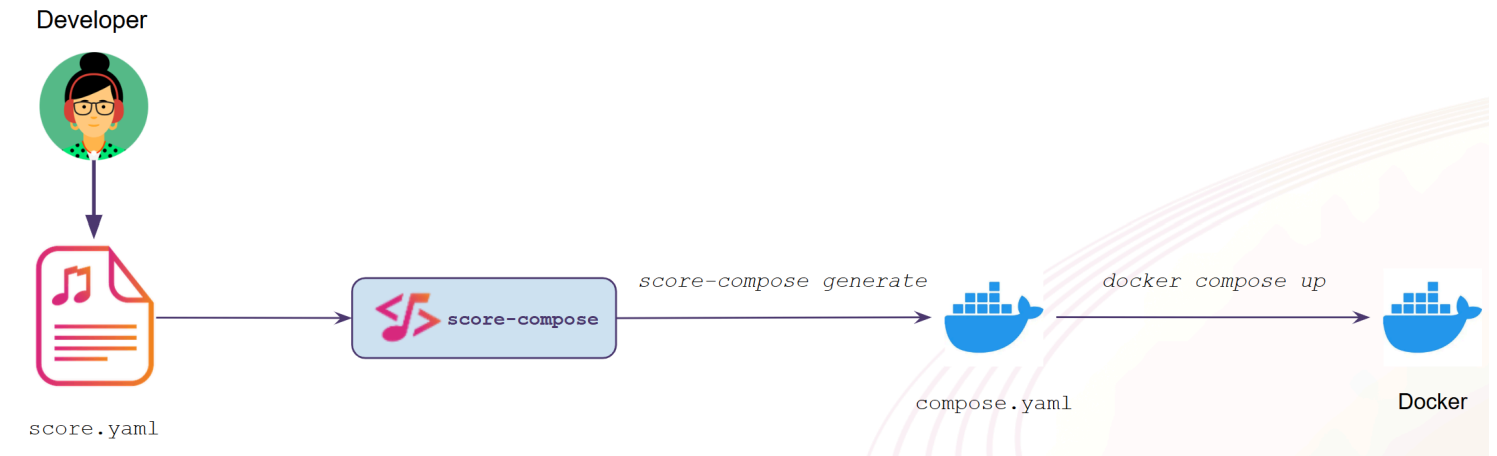
- [Score specification reference](#)

Getting started with **score-compose**

But from this **score.yaml** file we haven't deployed anything just yet.

Let's now deploy this workload locally with Docker Compose!

We'll use the **score-compose** implementation for that.



Initialize your current **score-compose** workspace, run the following command in your terminal:

```
score-compose init
```

*Note: The **init** command created the **.score-compose** directory to store a local state, read more about its purpose [there](#).*

Convert the **score.yaml** file into a deployable **compose.yaml**, run the following command in your terminal:

```
score-compose generate score.yaml \  
  --image scorespec/sample-score-app:latest
```

A new **compose.yaml** file has been generated, see [here](#).

All of this technical details abstracted by the **score-compose** implementation from the Developer.

Use Docker Compose to locally spin up our workload and its dependencies:

```
docker compose up -d
```

See the running containers:

```
docker ps
```

Test the deployed workload on [localhost:8080](#).

At this stage, we used two resource types: `postgres`, `dns` and `route`.

The available resource types and their inputs/outputs could be discovered like this:

```
score-compose provisioners list
```

We'll explore more about these resource types and provisioners later.

Resources

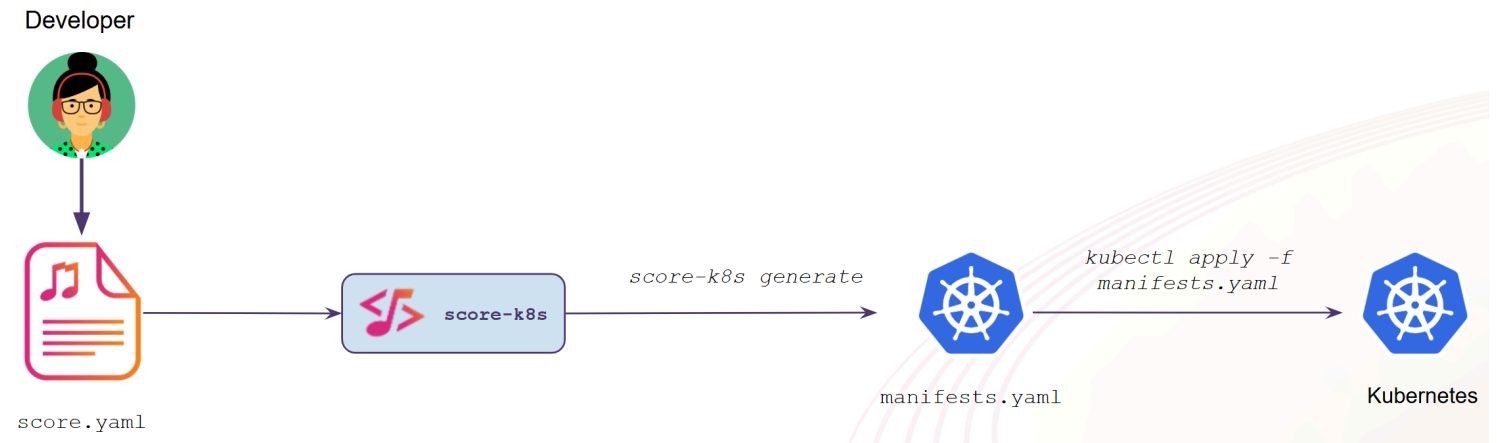
- [score-compose implementation](#)
- [Local state directory](#)

Getting started with **score-k8s**

From this exact same **score.yaml** file we now want to deploy it to Kubernetes.

Let's do it!

We'll use the **score-k8s** implementation for that.



Initialize your current **score-k8s** workspace, run the following command in your terminal:

```
score-k8s init
```

Note: The **init** command created the **.score-k8s** directory to store a local state, read more about its purpose [there](#).

Convert the **score.yaml** file into a deployable **manifests.yaml**, run the following command in your terminal:

```
score-k8s generate score.yaml \  
  --image scorespec/sample-score-app:latest
```

A new **manifests.yaml** file has been generated, see [here](#).

All of this technical details abstracted by the **score-k8s** implementation from the Developer.

Before deploying this **manifests.yaml** to Kubernetes, we need to cluster, create a local **kind** cluster by running this command:

```
sudo ./scripts/setup-kind-cluster.sh
```

Deploy these Kubernetes manifests to spin up our workload and its dependencies in Kubernetes cluster:

```
sudo kubectl apply -f manifests.yaml
```

See the running containers and associated resources:

```
sudo kubectl get all,statefulset,secret,httproute
```

Test the deployed workload on localhost:8080.

At this stage, we used two resource types: `postgres`, `dns` and `route`.

The available resource types and their inputs/outputs could be discovered like this:

```
score-k8s provisioners list
```

We'll explore more about these resource provisioners later.

Resources

- [score-k8s implementation](#)
- [Local state directory](#)

Advanced scenario with **score-compose**

The **score-compose** implementation CLI provides more options to enrich the default **compose.yaml** file generated in order to support more native Docker Compose features.

--build

Build the container image while deploying it via the **docker compose up --build** command, by using the **score-compose generate --build** feature:

```
score-compose generate score.yaml \  
  --build 'hello-world={"context":".", "tags":["hello-world"]}'
```

```
docker compose up --build -d
```

--override-property

Override any properties of the Score file imperatively:

```
score-compose generate score.yaml \  
  --image scorespec/sample-score-app:latest \  
  --override-property containers.hello-world.variables.MESSAGE="Hello, Compose!"
```

See the new **MESSAGE** value in the generated [compose.yaml](#) file.

--patch-templates

Override the default content and any fields of the **compose.yaml** with the patch template feature:

```
echo '{{ range $name, $spec := .Workloads }}  
{{ range $cname, $_ := $spec.containers }}  
- op: set  
  path: services.{{ $name }}-{{ $cname }}.read_only  
  value: true  
- op: set  
  path: services.{{ $name }}-{{ $cname }}.user
```

```
    value: "65532"
  - op: set
    path: services.{{ $name }}-{{ $cname }}.cap_drop
    value: ["ALL"]
  {{ end }}
  {{ end }}' | score-compose init --patch-templates -
```

In this example, this inlined snippet patches all the Workloads and sets the fields `read_only`, `user` and `cap_drop`.

```
score-compose generate score.yaml \
  --image scorespec/sample-score-app:latest
```

See the associated new fields and values in the generated [compose.yaml](#) file.

Instead of using the inlined format, you can use `--patch-templates` with local or external files (Https, Git, OCI), like this:

```
score-compose init \
  --patch-templates https://raw.githubusercontent.com/score-spec/community-
  patchers/refs/heads/main/score-compose/unprivileged.tpl
```

Resources

- [score-compose CLI](#)
- [Patch templates](#)
- [Patch templates examples Hub](#)

Advanced scenario with **score-k8s**

The **score-k8s** implementation CLI provides more options to enrich the default **manifests.yaml** file generated in order to support more native Kubernetes features.

--format yaml|kyaml

Export the **manifests.yaml** as [kyaml](#) instead of default **yaml**:

```
score-k8s generate score.yaml \  
  --image scorespec/sample-score-app:latest \  
  --format kyaml
```

See the new format of the generated [manifests.yaml](#) file.

--override-property

Override any properties of the Score file imperatively:

```
score-k8s generate score.yaml \  
  --image scorespec/sample-score-app:latest \  
  --override-property containers.hello-world.variables.MESSAGE="Hello, Kubernetes!"
```

See the new **MESSAGE** value in the generated [manifests.yaml](#) file.

--namespace

Set the **namespace** field for the generated Kubernetes resources:

```
score-k8s generate score.yaml \  
  --image scorespec/sample-score-app:latest \  
  --namespace test
```

See the new **namespace** field in the generated [manifests.yaml](#) file.

--generate-namespace

Generate a **Namespace** Kubernetes resource:

```
score-k8s generate score.yaml \  
  --image scorespec/sample-score-app:latest \  
  --namespace test \  
  --generate-namespace
```

See the new **Namespace** resource in the generated [manifests.yaml](#) file.

--patch-templates

Override the default content and any fields of the **manifests.yaml** with the patch template feature:

```
echo '{{ range $i, $m := .Manifests }}  
{{ if eq $m.kind "Deployment" }}  
- op: set  
  path: {{ $i }}.spec.template.spec.automountServiceAccountToken  
  value: false  
- op: set  
  path: {{ $i }}.spec.template.spec.securityContext  
  value:  
    fsGroup: 65532  
    runAsGroup: 65532  
    runAsNonRoot: true  
    runAsUser: 65532  
    seccompProfile:  
      type: "RuntimeDefault"  
{{ range $cname, $_ := $m.spec.template.spec.containers }}  
- op: set  
  path: {{ $i }}.spec.template.spec.containers.{{ $cname }}.securityContext  
  value:  
    allowPrivilegeEscalation: false  
    privileged: false  
    readOnlyRootFilesystem: true  
    capabilities:  
      drop:  
        - ALL  
{{ end }}  
{{ end }}  
{{ end }}' | score-k8s init --patch-templates -
```

In this example, this inlined snippet patches all the Workloads and sets the **securityContext** fields.

```
score-k8s generate score.yaml \  
  --image scorespec/sample-score-app:latest
```

See the associated new fields and values in the generated [manifests.yaml](#) file.

Instead of using the inlined format, you can use `--patch-templates` with local or external files (Https, Git, OCI), like this:

```
score-k8s init \  
  --patch-templates https://raw.githubusercontent.com/score-spec/community-  
patchers/refs/heads/main/score-k8s/unprivileged.tpl \  
  --patch-templates https://raw.githubusercontent.com/score-spec/community-  
patchers/refs/heads/main/score-k8s/service-account.tpl
```

In this example above, in addition to set the `securityContext` with this [unprivileged.tpl](#) patch template file, we are also using a second [service-account.tpl](#) patch template file to create a dedicated `ServiceAccount` resource per `Deployment`.

Provisioners

```
score-k8s init \  
  --provisioners https://raw.githubusercontent.com/score-spec/community-  
provisioners/refs/heads/main/route/score-k8s/10-shared-gateway-httproue.provisioners.yaml
```

```
score-k8s generate score.yaml
```

```
kubect1 apply -f manifests.yaml
```

Resources

- [score-k8s CLI](#)
- [Patch templates](#)
- [Patch templates examples Hub](#)

Conclusion

You've completed the **Score Labspace**! 🎉

✅ You now know:

- What is Score (Spec versus Implementations)
- How to author a Score file as Developers
- How to use the two default Score implementations: `score-compose` and `score-k8s`
- How to customize the outputs of the Score implementations with the patch templates feature
- How to define your own resource types with the resource provisioners feature

🎊 Well done!

Next Steps

- [Explore more examples](#)
- [CI/CD pipelines with Score](#)
- [Join the Score community](#)
- Any suggestions to improve this **Score Labspace**? Please help us make it better by filing an issue [there](#)!

Let's Score it! 🎵